

MS-GAN: GAN-based Semantic Segmentation of Multiple Sclerosis Lesions in Brain Magnetic Resonance Imaging

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Abstract—Automated segmentation of multiple sclerosis (MS) lesions in brain imaging is challenging due to the high variability in lesion characteristics. Based on the generative adversarial network (GAN), we propose a semantic segmentation framework MS-GAN to localize MS lesions in multimodal brain magnetic resonance imaging (MRI), which consists of one multimodal encoder-decoder generator G and multiple discriminators D corresponding to the multiple input modalities. For the design of the generator, we adopt an encoder-decoder deep learning architecture with bypass of spatial information from encoder to the corresponding decoder, which helps to reduce the network parameters while improving the localization performance. Our generator is also designed to integrate multimodal imaging data in end-to-end learning with multi-path encoding and cross-modality fusion. An additional classification-related constraint is proposed for the adversarial training process of the GAN model, with the aim of alleviating the hard-to-converge issue in classification-based image-to-image translation problems. For evaluation, we collected a database of 126 cases from patients with relapsing MS. We also experimented with other semantic segmentation models as well as patch-based deep learning methods for performance comparison. The results show that our method provides more accurate segmentation than the state-of-the-art techniques.

Index Terms—Semantic Segmentation, Image-to-image Translation, GAN, Multiple Sclerosis

I. INTRODUCTION

Multiple sclerosis (MS) is an inflammatory disease of the central nervous system. Magnetic resonance imaging (MRI) typically shows focal cerebral lesions primarily in the white matter of brain, and is the principal diagnostic. Quantitative analysis of the number and volume of lesions is essential during disease diagnosis and management, and accurate lesion segmentation is a fundamental step to facilitate such analysis. However, automated segmentation can be particularly challenging since MS lesions exhibit large variation in intensity, size, shape, and location. As a result, popular algorithms as U-Net [1] and GCN [2] would output over- or under-segmented

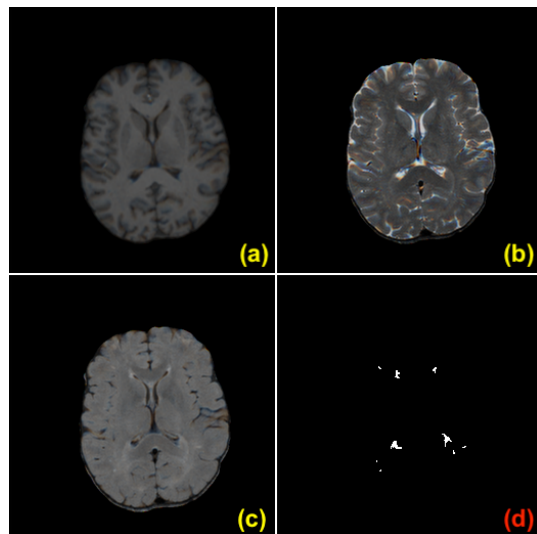


Fig. 1. Diverse information encoded in multiple modalities of brain MRI scans with skull removed for MS segmentation problem. (a) T1-weighted MRI. (b) T2-weighted MRI. (c) FLAIR MRI. (d) MS lesions ground truth. Since T1-weighted MRI show most clearly the boundaries between different tissues, anatomical structure can be captured in detail by T1-weighted MRI, while T2-weighted and FLAIR MRI scans are more sensitive to the pathological changes.

masks for MS lesions, as shown in Fig. 2. Although this issue can be alleviated by adjusting the weight in the cross entropy loss function, finding an appropriate value for the weight needs a group of empirical experiments and this weight setting will vary as the dataset changes.

In this study, we designed a new GAN-based semantic segmentation framework for MS lesions. and we make the following contributions: (1) *We adopt the LinkNet [3] as our base segmentation network.* LinkNet is an encoder-decoder network with additional linking from the encoder to the

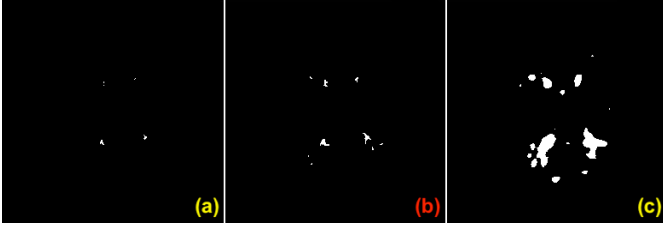


Fig. 2. Under- and over-segmentation of the MS lesions shown in Fig. 1. (a) Under-segmentation occurred when the cross-entropy weight was set too high. (b) MS lesions ground truth. (c) Over-segmentation occurred when the weight was set too low. To perform a reasonably good segmentation for MS lesions, a group of empirical experiments is necessary to find an appropriate value for the cross-entropy weight.

corresponding decoder to facilitate information sharing. The resultant LinkNet requires much fewer parameters than the other popular encoder-decoder networks (e.g. SegNet [4]), and hence is particularly useful in our problem domain with limited training data available. (2) *We modify the LinkNet structure to integrate the multimodal MRI data in an end-to-end learning architecture.* Diverse forms of MS-related information, including both anatomical structure and pathological changes, are encoded in different modalities of MRI separately, as demonstrated in Fig. 1. To process with this visual information, our model contains multiple encoding paths for multiple modalities, and the outputs from the multimodal encoders at each level are combined and linked to the corresponding decoder. (3) *We extend the conditional generative adversarial network (cGAN) to solve the semantic segmentation problem more efficiently by adding classification-related losses as regularization terms.* This extension has provided an alternative way to address under- and over-segmentation issues, which is typically achieved by adjusting the cross-entropy weight manually or applying morphological image filtering as an extra post-processing step. However, both approaches have their own drawbacks. The choice of morphological operations depends on the observation of the lesion masks generated, for example applying dilation and erosion for under- and over-segmentation, respectively. This requirement has limited the automation of segmentation algorithms. An appropriate cross-entropy weight needs to be found via empirical experiments and the setting will vary as the dataset changes. Our framework requires no empirical pre-experiments for fine-tuning the network model or extra post-processing step. (4) *We created a database of 126 MRI scans from patients with relapsing MS for evaluation.* Our experimental results show that our method outperforms other well-known deep learning models.

II. RELATED WORK

MS Lesion Segmentation Many MS segmentation techniques have been reported in the literature, such as those based on pixel-based classification [5], multi-atlas and dictionary learning [6], and graphical models [7]. More recently, the focus has shifted to deep learning, and, typically, patch-based convolutional neural network (CNN) models with sliding windows have been developed [8]–[10]. While the patch-based

approaches can increase the amount of data for training large networks, they cannot capture the spatial contexts in images and impose high computational cost for both training and testing stages.

To perform image-level segmentation efficiently, semantic segmentation models such as the fully convolutional network and encoder-decoder network have been proposed and incorporated into biomedical imaging [1], [11]–[13]. Such models can leverage the global spatial information in an image and output pixel-wise classification probabilities directly. In brain imaging, semantic segmentation methods have been designed and customized for glioma segmentation. For example, special layers are incorporated to effectively combine the multimodal imaging data [14], [15]. Additional paths are included to explicitly optimize the classification at the boundary area [16], [17]. The approach of semantic segmentation for MS lesions has been relatively uncommon. One study demonstrated the effectiveness of integrating semi-supervised learning with fully convolutional network to address issues with small training datasets [18]. However, due to the focus on semi-supervised learning, the segmentation performance is not optimized.

Image-to-image Translation and Generative Adversarial Networks (GANs) The supporting idea to achieve image-to-image translation is that the input and output images can be considered as different surface representations rendered from the same latent structure [19]. Since all input modalities should contain visual clues for revealing the output lesion mask, MS lesion segmentation problem can be formulated as an image-to-image translation problem, whose key is to learn a pixel-wise mapping function from inputs to outputs. Previously, these mapping functions were usually engineered by hand and this engineering progress generally relies upon specialized domain knowledge. Fortunately, the development of GANs tackles this issue by training the generator G and the discriminator D adversarially to approximate the mapping function.

Among all GAN-related variants [20]–[23], pix2pix can be seen as one of the most successful milestones reached in the domain of image-to-image translation problem [19], and its objective can be expressed as

$$L_{pix2pix}(G, D) = L_{cGAN}(G, D) + \lambda L_{L1}(G) \quad (1)$$

where $L_{cGAN}(G, D)$ is the loss of conditional GAN and $L_{L1}(G)$ is the $L1$ distance between the generated mask and the ground truth. In the optimization process of $L_{pix2pix}(G, D)$, the goal of G and D is to minimize and maximize it respectively, since the generator is tasked to fool the discriminator by producing a mask as close to the ground truth as possible, against the detection work performed by the discriminator to tell the difference between the generated mask and ground truth. $L1$ distance works as a regularization term added on this adversarial training progress.

This design of $L_{pix2pix}(G, D)$ has succeeded in approximating the mapping functions for many image-to-image translation problems [24]–[26]. However, we found that it could not

achieve a similarly accurate result for the semantic segmentation of MS lesions. We believe that this is primarily attributable to their design of GAN loss, which formulates the image-to-image translation problems as regression tasks, while semantic segmentation is intrinsically a dense classification task performed at the pixel-level. We therefore reformulated the problem by incorporating the classification-related objective term, namely cross-entropy loss and dice loss, into the design of our MS-GAN loss. This loss improvement alleviates the hard-to-converge issue that characterizes MS lesion semantic segmentation.

III. METHOD

Our MS-GAN is designed to learn a mapping from image input with multiple modalities $MODs$ to lesion mask y , $G : MODs \rightarrow y$. Hence, one generator with multiple modalities is needed as the generative model to approximate the mapping function above. Meanwhile, we adopt multiple discriminators for our MS-GAN framework, where each discriminator corresponds to a single modality. Those discriminators D would be optimized to detect the difference between the “real” ground truth and “fake” lesion masks produced by the generator G , against the adversarial learning process of the generator G where G is trained to output lesions masks that appear as real as possible. This optimization process is illustrated in Fig. 3.

A. Objective

The objective of our MS-GAN can be expressed as

$$L_{MS-GAN}(G, D) = L_{cGAN}(G, D) + \lambda[L_{L1}(G) + L_{SemSeg}(G)] \quad (2)$$

where $L_{cGAN}(G, D)$ and $L_{L1}(G)$ follow the same definition specified in Equation 1, while $L_{SemSeg}(G)$ is the additional loss related to the semantic segmentation task. This new regularization term $L_{SemSeg}(G)$ should encourage our MS-GAN to formulate this image-to-image translation problem closer to a classification task, since the nature of semantic segmentation is a pixel-wise classification task, rather than a regression work. Here, $L_{SemSeg}(G)$ is defined as

$$L_{SemSeg}(G) = L_{CrossEntropy}(G) + L_{Dice}(G) \quad (3)$$

where $L_{CrossEntropy}(G)$ and $L_{Dice}(G)$ are the cross entropy loss and the dice loss calculated for our generator respectively. Hence, our generator with the additional constraint $L_{SemSeg}(G)$, would prefer to produce a reasonable segmentation mask in the process of adversarial training against the discriminator D . We observe that this preference is significantly beneficial to the semantic segmentation problem, especially when the algorithm is trapped at a local minimum and hence the balancing point between G and D cannot step further easily.

The final goal of our framework is

$$G^* = \arg \min_G \max_D L_{cGAN}(G, D) + \lambda[L_{L1}(G) + L_{SemSeg}(G)] \quad (4)$$

B. Multimodal Encoder-Decoder Generator

Our generator model design is illustrated in Fig. 4. The main components in the model are the M encoding paths and one decoding path, with M indicating the number of imaging modalities. All encoding paths have identical structures, each consisting of four encoder blocks with residual connections within each block. The encoders could encode the image features with incremental downsampling, and the corresponding decoders based on full convolutions gradually upsample the feature maps to produce an image-level mapping at the output.

During the downsampling process, some spatial information within the images would be lost. To recover the spatial information on the decoder side, bypass connections are added from the encoder blocks to the corresponding decoders. In this way, information is shared between the encoders and decoders, and also this enables the use of fewer parameters in the decoder blocks. In addition, with multiple encoding paths for the multiple modalities, the encoder outputs at each level are combined together before being linked to the decoders, so that multimodal information is fused within the end-to-end learning. We experimented with various combination techniques, including sum, multiply, max pooling, and adding additional cross modalities convolution blocks [15] to merge the data via convolution. It is found that a simple sum across modalities produced the best result. Compared to the convolution approach, the sum operation could avoid extra parameter learning and hence lower the chance of overfitting.

The detailed network architecture including kernel sizes and feature map configurations can also be seen in Fig. 4. Prior to the encoders, images are passed through a convolution block with max pooling for an initial transformation. The designs of the encoders and decoders are based on the LinkNet model, and the encoding path follows the ResNet18 structure with weights initialized using the ImageNet-pretrained model [3]. After the decoders, three more convolution / full convolution blocks are added to generate the segmentation map at the image level. Batch normalization and ReLU are also added between the convolutional layers. The cross entropy loss $L_{CrossEntropy}(G)$ is used as the final loss layer.

C. Discriminator

We adopted PatchGAN as our base model for the discriminator, which would try to classify whether a given $N \times N$ patch is real or fake. N is set to 50 in our experiments. We argue that the visual features used to distinguish patches collected from different input modalities, should differ from each other. Hence, each modality would construct its own discriminator to focus on the feature extraction, which is uniquely representative for that specific modality only.

IV. EXPERIMENTS AND RESULTS

We evaluated the segmentation results with the voxel-level Dice coefficient, object-level recall, precision and F-score, and slice-level Hausdorff distance (HD). The object-level true positive was determined by a minimum 50% overlap between the segmentation output and ground truth.

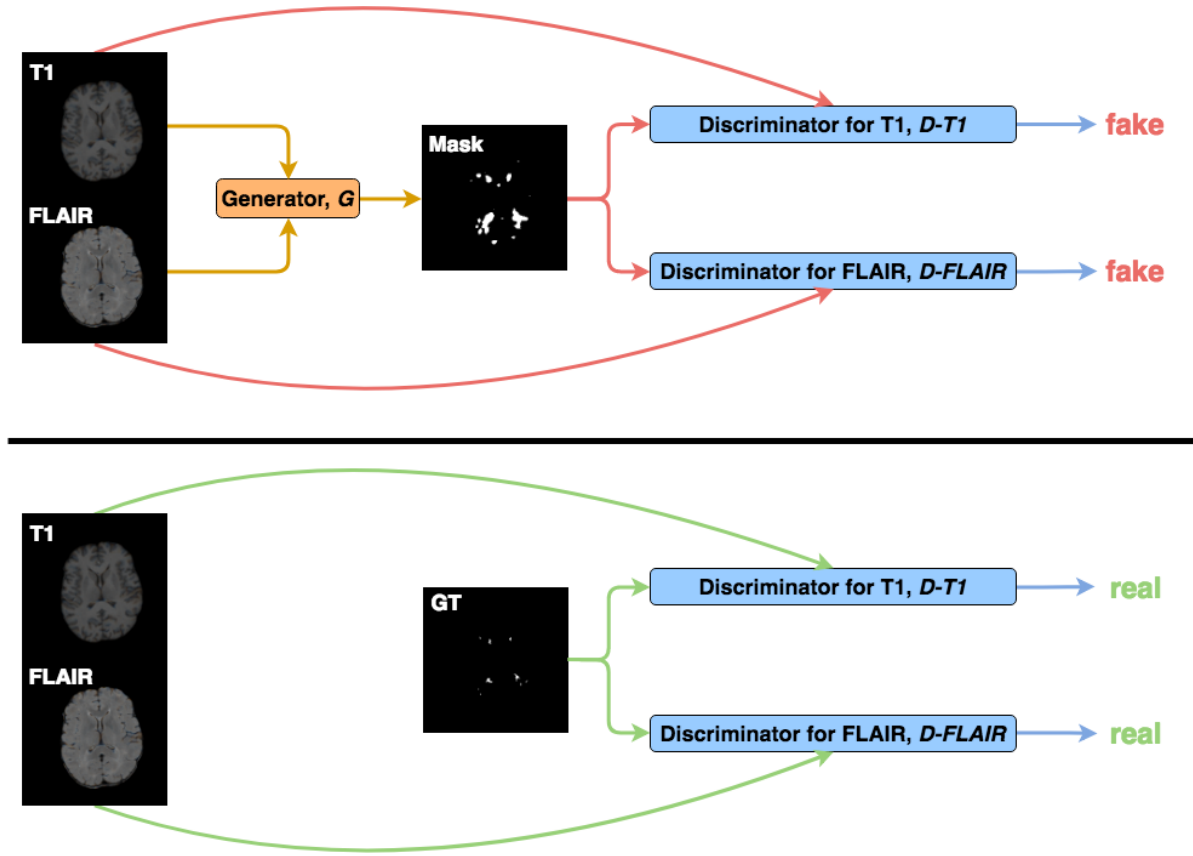


Fig. 3. The training process of MS-GAN to map from multiple modalities *MODs* to lesion mask *MASK*. Each modality has its own discriminator, D , to distinguish real (ground truth, modality) from fake (mask generated, modality) pairs. The generator, G , is not only targeted at tricking the discriminators, but is also expected to produce semantic segmentation for MS lesions.

A. Dataset and Implementation

A total of 69 consecutive patients with relapsing-remitting MS were included in this study (male / female: 33 / 36, mean age: 43.3 ± 11.2). Both baseline and follow-up images were acquired, producing a total of 126 scans. A 3T GE Discovery MR750 scanner was used to collect the T1-weighted MRI and FLAIR MRI, which are sensitive to the anatomical structures and pathological changes respectively, and the MRI protocol included: Sagittal 3D T1 (TE/TR/TI = 2.7/7.2/450 ms, voxel size = $1 \times 1 \times 1 \text{ mm}^3$), and FLAIR CUBE (TE/TR = 163/8000 ms, voxel size = $0.47 \times 0.47 \times 1.2 \text{ mm}^3$). T1 images were corrected for intensity inhomogeneity using the FreeSurfer N3 correction method. FLAIR and T1 were co-registered using FSL FLIRT. For ground truth annotation, lesions were identified on the co-registered FLAIR and T1 images and semi-automatically segmented using JIM 7 software (Xinapse Systems, Essex, UK) by a trained image analyst. The baseline lesion mask and follow-up lesion mask were generated independently. For evaluation, the patients were divided into two groups. The first 9 patients were selected to form the test group, resulting in 20 test volumes (2-3 scans per patient). MRI data of the other 60 patients were used as the training data (1-3 scans per patient), resulting in 106 training volumes (1-3 scans per patient).

Our model was implemented with the PyTorch backend on a standard PC with an NVIDIA GeForce GTX 1070 GPU. Both T1 and FLAIR slices were padded and resized to 512×512 pixels. Data augmentation was implemented with random horizontal or vertical flipping, and rotation of 90, 180, or 270 degrees. The Adam optimizer was used with a batch size of 16, an initial learning rate of 0.0002. The coefficient λ from our objective Equation 2 was set to 100, as suggested by [19]. The optimization of our MS-GAN framework could reach convergence in about 300 epochs.

B. Pseudo 3D slices

The multimodal encoder-decoder generator can be applied to MS lesion segmentation with *pseudo* 3D slices, which can be visualized in Fig. 5. Since an axial slice is a grayscale image, we treat one slice as one color channel in the RGB space and stack three consecutive slices together into one RGB image. The resultant color image becomes a pseudo 3D slice capturing some depth information. The network model is then applied to these pseudo 3D slices to perform semantic segmentation, and training is performed based on the binary segmentation mask corresponding to the middle slice.

Preparing pseudo 3D slices for our 2D segmentation generator can be considered as an efficient approach to learn the spatial information in sequence along the z-axis. An alternative

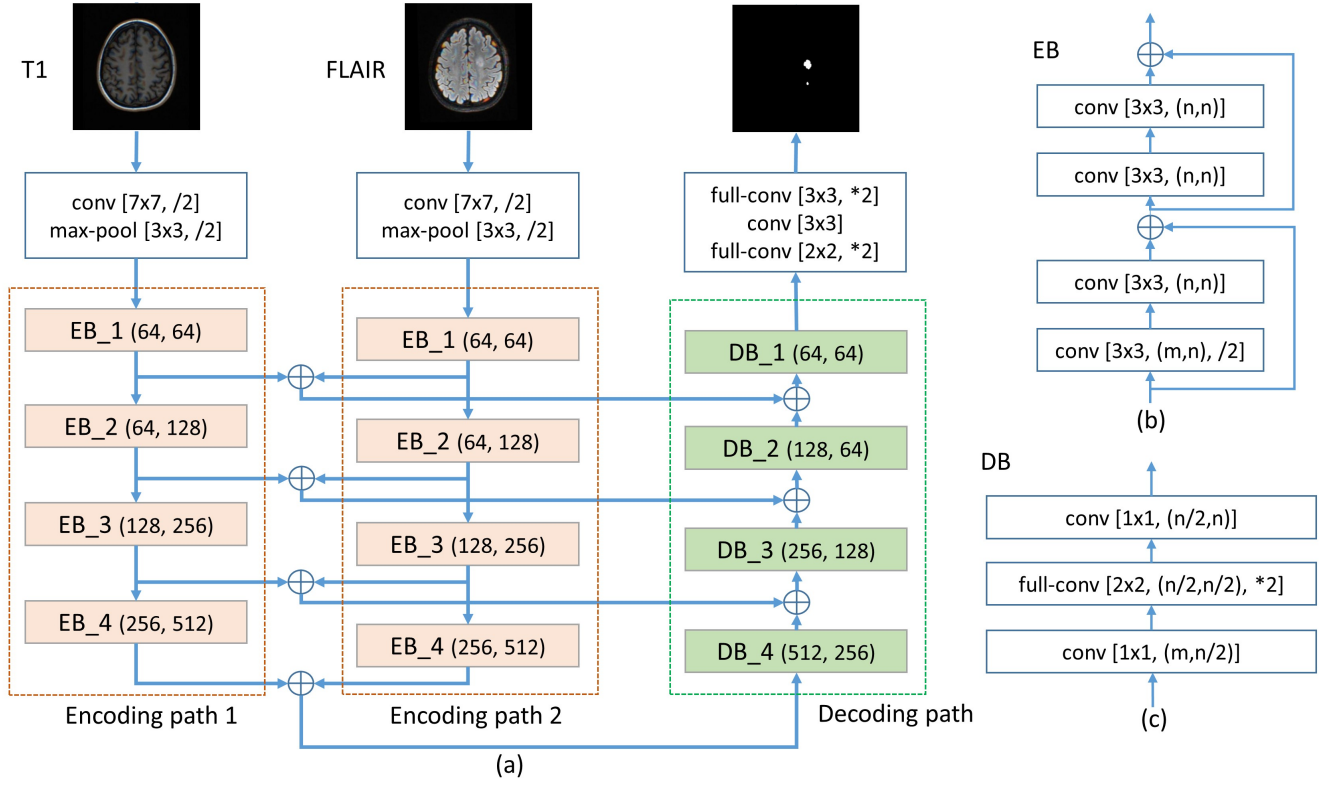


Fig. 4. (a) Visualization of the proposed multimodal encoder-decoder generator, which consists of two encoding paths for two imaging modalities (T1 and FLAIR) and one decoding path to generate the segmentation. EB and DB denote encoder and decoder blocks, respectively. The numbers in brackets are the m and n parameters used to control the input and output feature map sizes, as used in (b) and (c). (b) The structure of an encoder block with residual connections. (c) The structure of a decoder block. In (b) and (c), the kernel sizes and (input, output) feature map channels are specified, and /2 and *2 denote down- and up-sampling.

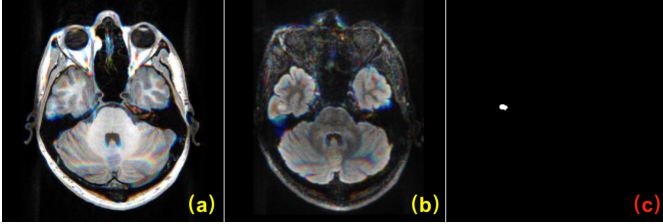


Fig. 5. Pseudo 3D slices encoding depth information as RGB channels. (a) The pseudo 3D slices for T1-weighted MRI. (b) The pseudo 3D slices for FLAIR MRI. (c) The MS lesions mask corresponding to the middle slice.

method would be treating the entire scan as a 3D tensor and adopting a 3D segmentation network to process it [27]. To make a comparison between these two approaches, we tried upgrading our network to a 3D version by replacing the 2D version of convolutional layer, MaxPooling layer, and batch normalization layer with the corresponding 3D version of them. The result, which can be viewed in (4) of Table I, indicates the fact that adopting a 2D network with pseudo 3D slices outperforms that of 3D network with 3D tensors. This may be because when using 3D network to process 3D tensors directly, the significant decrease in the number of training cases offsets the potential benefits gained from preserving all 3D spatial information. All following experiments adopt the

TABLE I
SEGMENTATION RESULTS EVALUATING OUR GENERATOR COMPONENTS.

	Dice	Recall	Precision	F-score	HD
Our generator	0.642	0.707	0.618	0.660	2.26
(1) T1 only	0.191	0.459	0.286	0.352	6.75
(2) FLAIR only	0.613	0.628	0.569	0.597	2.07
(3) Concatenation	0.623	0.605	0.650	0.627	2.36
(4) 3D network	0.566	0.472	0.519	0.494	2.31

2D segmentation network and use pseudo 3D slices as inputs.

C. Cross-Modality Fusion Components of Generator

To evaluate the effects of the cross-modality fusion design of our generator, we compared the following variations of our method: (1) T1 only, using only T1 images for segmentation; (2) FLAIR only, using only FLAIR images for segmentation; (3) modal concatenation, which uses the concatenated T1 and FLAIR images as network input without the multimodal encoding. To compare the performance difference caused by these variants of generator, we adopt them as segmentation networks and train them to output the MS lesions masks directly, without taking any advantage of the adversarial training process. None of the discriminators are applied to these experiments.

Table I shows the segmentation results using various approaches. Comparing the results of T1 only, FLAIR only, and concatenation, it can be seen that T1 alone does not provide sufficient information to identify MS lesions. FLAIR alone produced the lowest HD, since FLAIR shows the best visual localization of MS lesions without the influence of T1 at the lesion boundary. However, the combination of T1 and FLAIR improves the object-level performance with fewer false positives and hence higher precision. Our model is more effective than using concatenation, indicating the advantage of having multimodal encoding in our encoder-decoder generator.

D. Under- / Over-segmentation

As illustrated in Fig. 2, one of the most challenging issues for the MS lesions segmentation task would be the under- or over-segmentation. To address this issue, one most commonly used approach would be to hand-engineer the mapping function by adjusting the weight in the cross-entropy function for the segmentation networks. A relatively reasonable setting for the weight is the ratio between the total number of pixels belonging to the foreground objects and background objects, as represented by

$$Weight = \frac{N_{lesions}}{N_{non-lesions}} \quad (5)$$

where $N_{lesions}$ and $N_{non-lesions}$ represent the total size of MS lesions area and non-lesions area respectively. However, we observed in experiments that this setting does not work for the semantic segmentation of MS lesions at all, due to the large variation of MS lesions in intensity, size, shape, and location. Instead, before these popular segmentation networks [1]–[3] can be used to segment valid MS lesions, a group of empirical pre-experiments is essential to find an appropriate value for the weight, which would require extensive and iterative training of the deep learning models.

Our MS-GAN can address the under- or over-segmentation issues efficiently, by approximating the hidden mapping function through the adversarial training process, which is also the common advantage of other GAN variants [19]. This benefit can be achieved by using the value calculated by Equation 5 directly to set up the cross-entropy term defined in $L_{SemSeg}(G)$ and no more empirical pre-experiments are therefore required. Fig. 6 illustrates the visual comparison between the lesions mask generated automatically by our MS-GAN framework and under- or over-segmentation of the same MS lesions produced by another popular segmentation network U-Net.

E. Classification-related Constraint

Despite the advantage mentioned above, GAN and its variants have their own drawbacks as well, which is the hard-to-converge challenge. To alleviate this issue for the image-to-image translation problem particularly, some approaches prefer to formulate this pixel-wise translation problem as a regression task and therefore add the regression-related losses, for example $L1$ and $L2$ distance, as additional constraints to

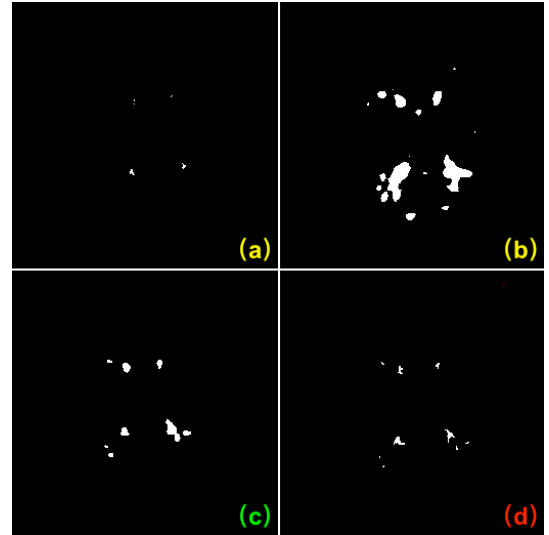


Fig. 6. Multiple Sclerosis lesions mask generated by our MS-GAN framework, compared to the under- and over-segmentation of the MS example shown in Fig. 1. (a) Under-segmentation appeared when the cross entropy weight is set as too high manually. (b) Over-segmentation happened when the weight is set as too low manually. (c) MS lesions mask generated automatically by our MS-GAN framework. (d) MS lesions ground truth.

TABLE II
SEGMENTATION RESULTS EVALUATING VARIOUS CONSTRAINTS.

	Dice
Generator only	0.642
(1) MS-GAN with L_{L1} removed	0.671
(2) MS-GAN with $L_{CrossEntropy}$ removed	0.317
(3) MS-GAN with L_{Dice} removed	0.665
(4) MS-GAN with all constraints preserved	0.672

the objective function of cGAN [19], [28]. These regression-related constraints have managed to provide a rough direction for the model to take the next step for the gradient descent, when a local minimum is reached in the adversarial optimization process of GAN. Nonetheless, we found that this guidance would work for some limited cases only, for example the image colorization problem, whose output images are expected to be the images of real-valued pixels, but not for the classification-based tasks, for example this MS lesions semantic segmentation challenge, whose target images consist of the discrete-valued pixels encoding for class information.

Inspired by this observation, we argued that an extra classification-related constraint $L_{CrossEntropy}$ is useful for this classification-based task. Since the Dice coefficient is commonly used for the evaluation of the semantic segmentation problems, we add L_{Dice} loss into the current constraint and combine them as $L_{SemSeg}(G)$. To evaluate the effects of those constraints, we compared the following losses used to train our MS-GAN against the Dice coefficient achieved by the generator directly: (1) MS-GAN with L_{L1} removed; (2) MS-GAN with $L_{CrossEntropy}$ removed; (3) MS-GAN with L_{Dice} removed; (4) MS-GAN with all constraints preserved.

Table II shows the segmentation results using various

TABLE III
SEGMENTATION RESULTS AND SIZE OF TRAINABLE PARAMETERS,
COMPARING OUR METHOD WITH OTHER DEEP LEARNING APPROACHES.

	Dice	Recall	Precision	F-score	HD	Params
Ours	0.672	0.692	0.724	0.708	2.19	29.1M
U-Net	0.405	0.376	0.343	0.359	3.99	2M
GCN	0.490	0.475	0.459	0.467	3.11	55.6M
DeepMedic	0.623	0.627	0.498	0.556	1.98	1M

approaches with different constraints added. The significant Dice drop with $L_{CrossEntropy}$ constraint removed indicates the appearance of the hard-to-converge issue resulting from attempting to use GAN to approximate a classification-based mapping function, which is formulated as a regression problem. The contributions made by L_{L1} and L_{Dice} can be viewed in Table II as well, so we decided to keep both of them. Furthermore, with the aim of achieving a similarly excellent performance for the classification-based image-to-image translation problems, we suggest putting an extra classification-related constraint on the adversarial optimization process of GAN and its variants.

F. Further Comparison

We also implemented several other well-known deep learning models for performance comparison, including: (1) U-Net [1], which is a fully convolutional network and has been widely used in biomedical imaging segmentation; (2) Global Convolutional Network (GCN) [2], which is a deep encoder-decoder network with global convolution blocks for better classification and boundary refinement blocks for improved localization; and (3) DeepMedic [29], which is a patch-based model with a customized 3D CNN architecture for brain lesion segmentation. For the compared approaches, we kept the original network structures with small changes to fit our imaging data. More complex data augmentation was applied to GCN and DeepMedic. Extensive configuration tuning was out of the scope of this study. We also compared the size of the network parameters.

Table III shows that our method largely outperforms the compared deep learning approaches. Although DeepMedic has the lowest HD, the HD values were only computed for true positive objects and the result would be biased by the lower recall of DeepMedic. Also, DeepMedic produced many false positive regions, hence the precision and F-score were low. U-Net is less complex than our model and GCN, and the segmentation performance was significantly lower than the other approaches. GCN contains the largest amount of parameters with the ResNet152-based encoders, global convolution, and boundary refinement blocks, and we implemented more extensive data augmentation to learn an effective GCN model. The performance advantage of our model over GCN demonstrates that a simpler LinkNet-based model can be more effective in modeling the brain MRI, which presents a limited amount of training data and tends to show less structural visual features compared to general images. Note that our model

contains extra discriminators networks for each individual modality inputted (29.1M corresponds to two discriminators situation) and hence the network parameters are relatively larger than U-Net and DeepMedic, but the more accurate segmentation results pay off in the end.

V. CONCLUSION

In this paper, we present a GAN-based semantic segmentation framework MS-GAN for multiple sclerosis in brain MRI. Our model consists of one multimodal encoder-decoder generator G and multiple patch-based discriminators D corresponding to the multiple input modalities. The multimodal design of our generator effectively integrates the different imaging modalities in the end-to-end learning, and the adversarial training process with the discriminators helps to refine the segmentation. Compared to other popular segmentation networks, our MS-GAN framework requires no empirical pre-experiments for fine-tuning the network or any extra post-processing step to address the under- or over-segmentation issue, which is one of the most difficult challenges in the domain of MS lesions segmentation. Further, we also define a classification-related constraint for the optimization of GAN, which helps to alleviate the hard-to-converge issue for the MS lesion segmentation task. Experiments were conducted on a set of 126 cases from clinical studies. The evaluation results show that our method is more effective than the other popular semantic segmentation models and a patch-based 3D CNN model that is customized for brain MRI.

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